

SIEMENS

Spectrum Power 5 ICCP Protocol

5.30

Technical Specification

Revision Record

Preface

Open Source Software

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NOTE

For your own safety, observe the warnings and safety instructions contained in this document, if available.

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Revision Record

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04-2-00	08.2006			SPC 4.2 version features updates
04-1-00	10.2004			Update for SPC 4.1

Preface

Notes on Safety

This manual is not a complete index of all safety measures required for operation of the equipment (module or device). However, it comprises important information that must be followed for personal safety, as well as to avoid material damage. Information is highlighted and illustrated as follows according to the degree of danger:



WARNING

WARNING means that death or severe injury **may** result if the measures specified are not taken.

- ✧ Comply with all instructions, in order to avoid death or severe injuries.



CAUTION

CAUTION means that medium-severe or slight injuries **can** occur if the specified measures are not taken.

- ✧ Comply with all instructions, in order to avoid moderate or minor injuries.



NOTE

Important information about the product, product handling or a certain section of the documentation which must be given particular attention.

Qualified Electrical Engineering Personnel

Only qualified electrical engineering personnel may commission and operate the equipment (module, device) described in this document. Qualified electrical engineering personnel in the sense of this manual are people who can demonstrate technical qualifications as electrical technicians. These persons may commission, isolate, ground and label devices, systems and circuits according to the standards of safety engineering. Only competent and authorized personnel should work with this product after becoming thoroughly familiar with all warnings, safety notices, operating instructions and maintenance procedures.

Proper Use

The product must not be used for any other purposes than that described in the technical documentation. If it is used together with third-party devices and components, these must be recommended or approved by Siemens AG. The successful and safe operation of this product is dependent on adequate transportation and proper handling, storage, installation, operation and maintenance.

Scope

This document provides specific and detailed information on how to use a particular product or product component.

**NOTE**

Note that the screenshots used in this document contain sample data which may not be available in some systems.

Typical Users

This document is designed for users that are already familiar with operational and technical aspects of power generation and power transmission and distribution as well as the product concepts:

- **System Engineers**
The System Engineer is able to install and to customize the system. He needs deep knowledge about the internal structures and processes of the network control system.
- **Data Engineers**
The Data Engineer translates the real network of a utility or the generation plants to the data model of the network control system. He also maintains the data model.
- **Technical Specialists**
The Technical Specialist is responsible for a special technical area which is related to the network control system. This can be the planning department, which plans changes in the network and uses network analyses applications of the network control system or the RTU department, which has to connect the RTU to the network control system. The Technical Specialist has to know how specific functions of the network control system work in detail.
- **Project Manager**
The Project Manager has to discuss technical issues with the customer. Therefore he needs a functional overview about the system and in some cases also specific technical specifications, which describe the way the system works. Those descriptions are also in some cases part of the contract.
- **Proposal Manager**
The Proposal Manager has to be able to read and understand the tender from the customer and to map the required functionality to the product and the way how the function can be fulfilled with the product. Depending on the tender this requires also the knowledge of a technical specialist. In addition the Proposal Manager needs an overview description of the system for the tender. In this description at least the functions, the tender is asking for, are described on a certain level.

Structure of the Manual

- **Introduction:**
Basic information about the component
- **Functionality:**
Description of different functionality of the component
- **Technology:**
Information about user interface technology and external data interfaces
- **Algorithm/Concepts (optional):**
Description of calculation algorithms and concepts

Documentation Conventions

Example	Meaning
File > Open	Selecting the menu item Open on the File menu
F1	Pressing the F1 function key on the keyboard
Control+o	Holding down the Ctrl key and pressing the letter o on the keyboard
Control+Shift+o	Holding down the Ctrl key and the Shift key and pressing the letter o on the keyboard
Left-click	Clicking the left mouse button

Example	Meaning
Right-click	Clicking the right mouse button
Shift-click	Holding down the Shift key on the keyboard and clicking the left mouse button

Questions?

This document is part of a set of manuals that describes the complete product.

Various other product modules may be mentioned or discussed in this document. For more detailed information or if you have any questions about these products, please contact your Siemens representative.

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1 Introduction

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1.1 Basics

The ICCP Protocol (ICCP) is designed to allow data exchange over a network (WAN or LAN) between a utility control center and the following:

- Other utilities
- Energy pools
- Regional control centers
- Non-utility generators

The data exchanged is primarily real-time system information.

For example, measured analog values, digital values, and accumulator values, along with supervisory control commands. Data exchange occurs between one SCADA or EMS host control center and another host control center, typically through a WAN. The WAN consists of a dedicated point-to-point communications interface, frame relay, or other internet-based transport mechanisms. The data exchange path (ICCP link) traverses one or more intervening communications processors.

For example, communication devices such as routers, bridges, modems, etc.

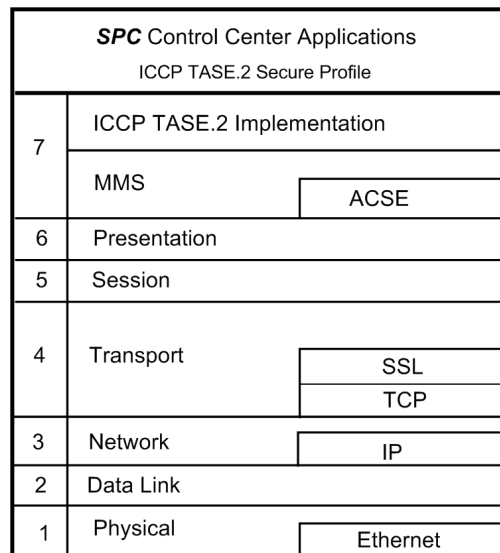
ICCP is designed to provide a communication protocol for use within the utility industry. It is an international standard, officially known as IEC 60870-6 TASE.2 protocol (TASE.2).

The ICCP standards are **companion standards** of the Manufacturing Message Specification (MMS) and ICCP is a user of MMS services.

ICCP performs the following functions:

- Provides a standard protocol to augment the existing (several) data exchange protocols.
- Simplifies the problems of data exchange between control centers.

The relationships of Spectrum Power 5, ICCP, MMS, and the remainder of the OSI stack are as shown in the following figure:



[ticip008-160411-01.tif, 1, en_US]

Figure 1-1 Protocol Relationship

ICCP ICCP protocol
MMS Manufacturing Message Specification
ACSE Application Control Service Element

ICCP is designed to operate using the OSI 7-layer reference model. The Spectrum Power 5 ICCP product supports the TCP/IP protocol in the lower layers. For ICCP to use TCP/IP in the OSI model, it uses recommendation RFC-1006.

1.2 ICCP Model

This model of a control center includes a number of classes of applications, which need not be present. An individual control center in its interactions with other control centers, acts as client, server, or both.

The control center acting as a server appears as a singular entity to clients, irrespective of the architecture of the system in use at a control center. A server control center interacts with multiple clients. This model of a control center consists of one or more processes. These processes perform the communications that allow control centers to acquire data, change data, and control devices. The ICCP specification defines the services and protocol necessary for performing the communication between these processes. The ICCP model defines a number of operations and actions. An operation is associated with an ICCP client. An action is associated with an ICCP server request and an immediate ICCP server response.

Bilateral Agreement

Rules and restrictions defined in a bilateral agreement govern the data exchange between control centers. These rules and restrictions restrict the clients view of a control center to a subset of the data that actually exists at that center.

The **Bilateral Agreement** specifies the following:

- Terms and conditions of the exchanges between the control centers, including the data accessible to each control center.
- The conditions under which the data is exchanged.
- The hosts that is used to establish the network connection.

Bilateral Table

The bilateral agreement is implemented within the software as a **Bilateral Table**. The **Bilateral Table** is a concept in the ICCP model. It is not necessarily a database table, nor any collection of data structures that support the bilateral agreement. In Spectrum Power 5, the **Bilateral Table** is implemented using the Information Model Management (IMM).

ICCP Associations

The ICCP association concept is based upon the MMS model. Associations occur between 2 ICCP instances. Associations are typically long lasting. They are established when the 2 instances initiate. They remain in place as long as the instances continue to run and the communication connections between them remain intact. ICCP instances normally remain running until an exception condition.

For example, offline maintenance, error, and operator shutdown occurs.

The 3 ICCP operations are defined for managing associations:

- **Associate** is used to initiate the establishment of an association.
The Spectrum Power 5 ICCP solution allows either the client or server to initiate the association.
- **Conclude** is used to complete an association.
Either the client or server use **Conclude**.
- **Cancel** is used to complete an association when failures have occurred in the underlying communications mechanisms.
Either the client or server use **Cancel**.



NOTE

ICCP standard indicates the **Associate** is an operation (thus performed only by the client) and the **Conclude** and **Cancel** are considered to be both operations and actions since either a server or client invoke them. Spectrum Power 5 treats the **Associate** as both an operation and an action, which means the server can also initiate the association. The Spectrum Power 5 solution allows a host to be both the client and server on the same association.

Some vendors do not allow a host to be both a client and a server on the same association. To allow interoperability, Spectrum Power 5 is configured to use 2 associations when communicating with such vendors. For one association, the local host is a pure client, for the other association, the local host is a pure server.

Application Reference (AR) Names

While not actually part of the ICCP standard model, the term **Application Reference Name** is used to identify **Application Entity** (AP Title and AE Qualifier) information and the presentation address associated with an ICCP application. The local AR specifies the local ICCP application instance, and the remote AR specifies a remote ICCP application instance.

In Spectrum Power 5 ICCP, a local AR name uniquely identifies the address of a process on the local system that supports the ICCP communication with a specific remote control center (RCC). Each RCC has associated remote AR names that specify an end point on that system. These AR names are part of the configuration.

Conformance Blocks

ICCP consists of a set of services referred to as **Conformance Blocks**. Each **Conformance Block** defines a specific set of functions that is deployed in the control center. Block 1 is the minimum level of support that an ICCP vendor offers. All remaining blocks are optional. Spectrum Power 5 ICCP currently supports blocks 1, 2, and 5.

The definitions of the blocks are:

- Block 1
Basic Services
- Block 2
Conditions (Extended Dataset Condition Monitoring)
- Block 5
Supervisory Control (Device Control)

2 Functionality

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2.1 Block 1 - Basic Services

This block function is used to exchange the real-time power system data between Spectrum Power 5 and the remote systems. It includes basic services. For example, services that are used to control the association, getting data values, setting up datasets, and periodic reporting of datasets.

Periodic reporting is configured to report all the requested values, or that have changed (report by exception). Different periods can be assigned to different values.

For example, some values are requested every 10 sec, and other values are requested every 5 min, or every day.

The Spectrum Power 5 ICCP organizes requested data with the same characteristics (including period, report by exception option, and priority) into structures called **Transfer Groups**. Standard ICCP exchanges structures called **Datasets**, has a maximum length. By defining transfer groups, and automatically mapping them into datasets, Spectrum Power 5 ICCP relieves the system or communication administrator to manage datasets and their sizes.

Client Functions

Based on association initiation, ICCP performs the following tasks for each transfer group defined in the configuration:

- Maps a transfer group into one or more datasets ensuring that no dataset exceeds the maximum message size for a given association.
- Assigns dataset names based on dataset contents.
- Checks whether the server retained datasets from a previous association and, if so, aligns them with the ones that are needed based on current client configuration. The client defines new ones that the server does not know about and deletes old ones that are not needed.
- Requests an initial set of data values by dataset name for each dataset within a transfer group. It is done to verify the availability of the configured data values.
- Requests a dataset transfer set for each dataset within a transfer group.
- Writes the dataset transfer set to start server condition monitoring as specified by the group definition in the client configuration.



NOTE

A dataset is defined only on the client side and the client automatically tells the server about the dataset definitions. No effort is required to coordinate the definitions of the datasets.

Server Functions

When processing the start transfer request from the client, the server starts timers to monitor various transfer conditions. When the timer expires, the ICCP server generates a transfer report for the dataset associated with that transfer set.

Transfer Set

An object used to control data exchange by associating data values with transmission parameters. For example, time intervals.

Transfer Conditions

The transfer conditions supported in block 1 are:

- IntervalTimeOut
- IntegrityTimeOut

The IntervalTimeOut and IntegrityTimeOut are used to trigger transfer reports.

Transfer Reports

A transfer report contains a dataset name and values. Depending upon the transfer sets, a report contains ICCP values or ICCP instance names and values. ICCP values contains actual values, quality, or time stamp.

The transfer report action is mapped into the MMS Information Report service. The Report by Exception (RBE) attribute, specifies the format of the report.

For IntervalTimeOut, the transfer report is generated based on the RBE attribute **TRUE** or **FALSE**. If RBE is set to **TRUE**, only the data objects changed since last report is included in the transfer report with their object names. Otherwise, all the data objects in the dataset are included in the transfer report.

The IntegrityTimeOut condition is used to trigger periodic transfer reports. It includes whole dataset that is an integrity scan when RBE is set to **TRUE**.



NOTE

The transfer report in the BlockData format (Conformance Block 3) is not supported.

2.2 Block 2 - Conditions

Block 2 supports the spontaneous (as opposed to periodic) exchange of real-time exception data between Spectrum Power 5 and the remote systems.

The client and server functions for block 2 are the same as for block 1 except the condition monitoring of the **ObjectChange**. The **ObjectChange** is used to trigger a transfer report whenever an object within the dataset changes.

ObjectChange Processing

As part of the TASE.2 2000-08 revision, a new capability was added to support the transmission of all changes that Siemens associates with **ObjectChange** processing. In previous revisions, when **BufferTime** expired, only the last value of a point was reported. With 2000-08, a new capability called **All Changes Reported** was introduced. If all changes reported is specified, the server is to report all value changes of the point that occurred during **BufferTime**.

The Spectrum Power 5 ICCP server handles object change processing in the following way:

On startup, the ICCP server uses the Spectrum Power 5 **Data Distribution Service** (DDS) to register for change notification with the Spectrum Power 5 application database. It is used for all the server data objects it is configured to provide. This allows it to support the **ObjectChange** condition if it is specified in a start transfer request from the client.

If **AllChangesReported** is true, the server starts to buffer changes once the client enables the transfer set. It stops when the client stops the transfer set or the association with the client is broken.

When it receives notification from Spectrum Power 5 that an object has changed and if no **BufferTime** is specified. The ICCP server generates the transfer report. If the **BufferTime** is non-zero, the ICCP server waits for the specified **BufferTime** before generating the transfer report. All data points that changed within the **BufferTime** are included in this transfer report. If there are multiple changes to a data point within this **BufferTime**, and **AllChangesReported** is set to false, only the last value is included in the transfer report. If there are multiple changes and **AllChangeReported** is set to true (and RBE is true), all the changes that occur to the data point during **BufferTime** are included in the transfer report.

The function of RBE attribute and the IntegrityTimeout transfer condition are the same as in block 1. They are used with the **ObjectChange** condition.



NOTE

The transfer report in BlockData format (Conformance Block 3) is not supported.

2.3 Additional Block 1 and 2 Client Feature

The Spectrum Power 5 implementation of ICCP includes a couple of additional features that is applied to the block 1 and 2 client data received from the remote. They are **Deadband** and **Scale and Offset**.

Deadband (Threshold)

When configuring block 1 and 2 client objects, it is possible to specify a **Deadband** attribute. When ICCP receives an incoming value from the remote system, it compares the new value with the previously received value. If the difference is greater than the **Deadband**, the value is passed to DDS. This is used primarily with analog measurements to prevent little changes in value from being propagated throughout the local system. A **Deadband** value is specified as an attribute for the remote. The remote attribute is used for client objects where no **Deadband** or a zero **Deadband** attribute was specified.

Scale (Slope) and Offset

Scale and **Offset** are also attributes that are specified for block 1 and 2 client objects.

When ICCP receives an incoming value from the remote system, it applies the formula:

$$Y = AX + B.$$

where,

A represents the Scale

B represents the Offset

X represents the incoming value.

The resultant value, Y, is the value that is sent to DDS and the rest of the system. The default value for Scale is 1, and Offset its 0. Using a Scale of -1, the sign of the incoming value can be inverted. By providing Scale and Offset, any linear type conversion can be performed.



NOTE

The **Deadband** is applied after the **Scale and Offset** conversion occurs.

2.4 Block 5 - Supervisory Control

After the association has become active, the ICCP protocol engine process can exchange client or server control requests. The Spectrum Power 5 supervisory control module issues control commands. As a result, device control messages are sent to a remote control center.

Block 5 controls received from the remote node are passed to the supervisory control subsystem for additional processing. Control actions initiated by an operator or application that are sent to the remote node comes from **Supervisory Control** subsystem.

In addition, to exchange control actions across ICCP, Spectrum Power 5 also supports the ability to use block 5 to set tags on the remote system.

For additional information regarding the transmission of tags across ICCP and how it was implemented, refer to [4.7 Spectrum Power 5 Interpretation of the TASE.2 Tags Definition](#)

Client Functions

An ICCP client gets the supervisory control notification from DDS. ICCP sends it to the remote node as a control action. ICCP subscribes to DDS for all client control points (points defined as block 5 client objects).

Server Functions

An ICCP server uses a supervisory interface to send control commands it receives from the remote node to the local supervisory control subsystem. These are for the server control points (defined as block 5 server objects).

2.5 Functionality Introduced with TASE.2 2000-08

TASE.2 2000-08 introduced some new functionality. The main changes are summarized here along with how Siemens Spectrum Power 5 use these capabilities. This function works when communicating with other remote control centers that support the 2000-08 revision of TASE.2

All Changes Reported

In the previous revision of TASE.2, when the **ObjectChange** processing was configured, the server is required to send the current value for the specific object, even if multiple changes occurred to that object during **BufferTime**.

In TASE.2 2000-08 revision, a new **AllChangesReported** setting is added to the DSTransferSet that the client writes to the server. If **AllChangesReported** is set to **TRUE**, the server is to buffer all the changes that occur to a data point during BufferTime. When it is time to report, it sends all the values for the data point. If it is set to **FALSE**, the server works as it did in previous revisions, just reporting the current value.

For a proper implementation of AllChangesReported in Spectrum Power 5, the following settings are completed on the client side:

- AllChangesReported - True
- BufferInterval - > 0
- RBE - True
- TransferType - Report by Event (ObjectChange)

Data Types That Include Millisecond Time Stamps

TASE.2 2000-08 defines 3 additional ICCP data types used for block 1 and 2. These new types are variations of the real, state, and discrete types with the addition of a time stamp that includes millisecond resolution (referred to as time stamp extended).

The types are as follows:

- Analog (Real) plus quality plus time stamp extended
- State plus quality plus time stamp extended
- Discrete plus quality plus time stamp extended

SOE Data Exchange

SOE is not defined in revision 2000-08. Siemens Spectrum Power 5 implementation has a solution for it as follows. The **Millisecond time stamps** and **AllChangesReported** capability are implemented so that it is used to send and receive **Sequence of Events** (SOE) data. On the server, you can indicate if the source of the value it sends to the remote is the Telemetered DDS event of the SOE DDS event. On the client, you can indicate if the received data is fired into SCADA as a Telemetered DDS event, an SOE DDS event, or both.

The standard implies that **AllChangesReported** only includes from when the exchange is enabled and until it goes down. So changes to values that occur while the association is down is not saved or reported by the server.

The data that is exchanged and treated as SOE uses ICCP data types that include millisecond time stamps.

Using Block 1 and 2 to Transmit Tags

The TASE.2 2000-08 adds the ability to report Tags using block 1 and 2 reporting mechanisms. Initially, TASE.2 only defined how to set tags using block 5 controls.

For additional information, regarding the transmission of tags across ICCP and how it is implemented in Spectrum Power 5, refer to [4.7 Spectrum Power 5 Interpretation of the TASE.2 Tags Definition](#).

2.6 Remote Fetch

TASE.2 defines operations that allow a client to get the data value names and types from the remote server (fetches the information from the remote). The information retrieved by these operations are used to assist in the configuration of the ICCP block 1 and 2 client information in IMM.

To perform this remote fetch, you have to configure in IMM, ICCP remote, and association information.

Hence, an association is established between the local and remote systems. On the local system, you want to configure the server objects this system serves to a remote workstation. This allows the remote workstation to fetch the server objects the local system is providing.

Once an association is established with the remote system, a database administrator can initiate the remote fetch process. The remote fetch process uses the same association that is used to exchange any previously configured data. When the remote fetch takes place, it impacts the normal data exchange, so the remote fetch execution requires interaction with the operator.

The results of the remote fetch from a remote are 2 .xml files that are ready to be imported into IMM:

- The first .xml file contains a pseudo station definition where, ICCP can dump the new client values that it receives from the remote.

The name of the pseudo station is the name of the remote in the ICCP configuration. The measurement names in the station are the same as the ICCP object names. The types for the measurement are derived from the object type.

- The second .xml file contains the ICCP definition of the ICCP block 1 and 2 client objects.

If this file is imported as is, the ICCP client object has links to the corresponding measurements in the pseudo station. It is desirable to modify this file before importing, changing the link on the ICCP object to a different measurement in a real station. Another change is to break the transfer group created into multiple transfer groups appropriate for the data being exchanged.

With the pseudo station and ICCP definition imported and activated, point out check can easily take place to verify that the values are coming across ICCP correctly. When appropriate, another IMM job is created and the ICCP client objects are linked to proper measurement in a true station.

The remote fetch (actually the get data value type operation) is not able to distinguish between 2 state types on the remote server. They are **State Value** and **State plus Quality**. When ICCP retrieves a type that could be either, the local system treats it as **State plus Quality**.

3 Technology

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3.1 Internal and External Interfaces

3.1.1 User Interface

The ICCP user interface consists of the ICCP subsystem, ICCP remotes, ICCP links, and ICCP transfer groups in a tabbed format. The ICCP user interface also presents the values of ICCP server and client data objects. Each tab has a list view to list all the attached detail status information. ICCP is invoked from the **Runtime Explorer** application.

The ICCP user interface allows the user to control data exchange with remote systems (remote control centers). The functions are:

- Enable and disable the ICCP.
- Connect and disconnect remote systems.
- Control the client transfer groups configured for a remote node.
- View association status with remote system and if association is secure or non secure.
- View the ICCP client object values for a specific client transfer group on a specific remote node.
- View the ICCP server object values for a specific remote node.

ICCP Main Display

The ICCP main display shows the ICCP subsystem overall administrative and operating state and the administrative and operating state for each ICCP remote. It provides the operator with the capability to enable or disable ICCP and to connect or disconnect with the listed remote systems.

It also provides the mechanism for initiating the remote fetch process.

ICCP Link Display

The ICCP link display shows the operating state of each ICCP link with the remote system.

ICCP Transfer Group Display

The ICCP transfer group display has only one list view control. It represents the administrative and operating state of transfer groups.

ICCP Server Data Display

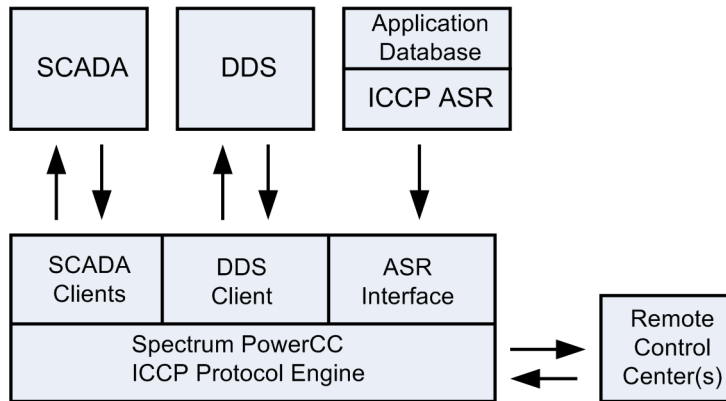
The ICCP server data display allows the user to view the ICCP server objects available to a remote system.

ICCP Client Data Display

The ICCP client data display allows the user to view the ICCP client data being received from the remote system. The client data presented is based on the client transfer group selected.

3.1.2 Application Interfaces

The ICCP protocol engine exchanges data with the remote ICCP system using physical connection.



[tsiccp01-150411-01.tif, 1, en_US]

Figure 3-1 Interface Overview

ICCP protocol engine acquires its configuration data from the ICCP ASR.

The ICCP protocol engine uses the **Data Distribution Service** (DDS) for the exchange of measured values between the remote system and the rest of Spectrum Power 5.

The ICCP protocol engine uses several SCADA clients to exchange information directly with SCADA. This includes clients such as the supervisory control client for incoming Block 5 control actions, and the marker client for the exchange of Tags.

3.2 System Characteristics

3.2.1 System Availability



NOTE

For additional information on Spectrum Power 5 system characteristics, refer to Technical Specification System Software.

Component Distribution

The ICCP protocol processes run on the ICCP servers.

The ICCP monitor process is configured to run on the RTS server. The purpose of this process is to mark the operating state of the remote control centers from UP to DOWN, if the ICCP servers go away unexpectedly. The monitor process accesses the ICCP ASR interface.

The Runtime Explorer uses an ICCP UI control to show ICCP-specific information. It is able to access the ICCP ASR.

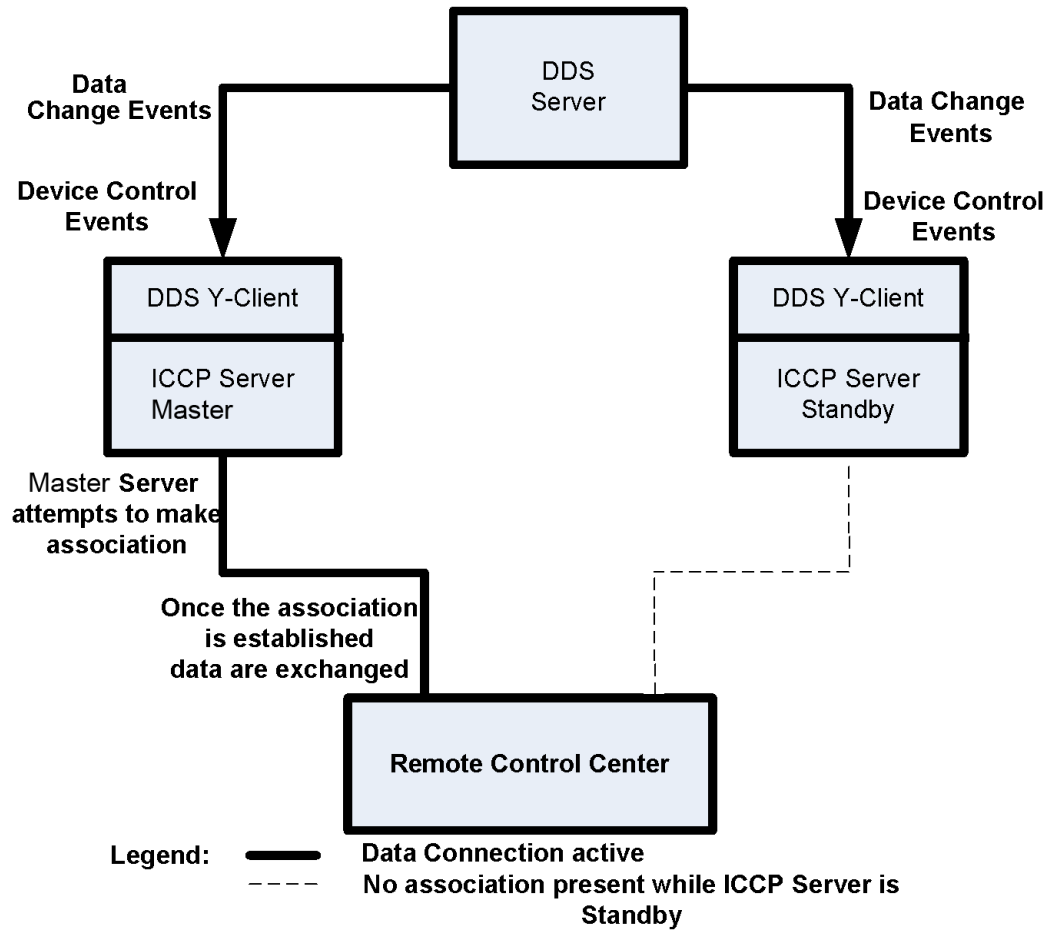
The **OSIFileCreator** is a tool that builds the configuration files required for the third party product. ICCP uses the lower layers of the protocol. It gets the information from IMM and writes the configuration files out to the ICCP servers. It normally runs on the PSOS.

Application Data Storage

The configuration of the ICCP is done within IMM. This configuration information is propagated to the ICCP ASR for use by the ICCP application. The ICCP application does not maintain any dynamic data in the ICCP ASR.

3.2.2 ICCP Server Redundancy

The ICCP redundancy design follows a **hot standby** strategy. Basically, it means that in a system with a redundant pair of ICCP servers, one is the active server and the other is the standby server. The servers base their **master/standby state** on the information received from the Startup/Shutdown/Recovery Service (SSR) using the ART interface.



[tsiccp03-241109-01.tif, 1, en_US]

Figure 3-2 Master or Standby Server Redundancy Concept

Master Server Functions

- Configures all internal data structures at startup time based on the ICCP configuration or domain data found in the ICCP ASR.
- Attempts to make an association with all remote control centers configured in the ICCP ASR.
- Receives all incoming data from the remote control centers.
- Uses the DDS Y-client interface to send all incoming data to the configured DDS server (most likely running on the real-time server).
- Receives all outgoing data change events (block 1 & 2) using the DDS server and sends this data to the corresponding remote control center.
- Receives all outgoing device control events (block 5) using the DDS server and sends these control requests to the corresponding remote control center.
- Listens on the ART callback interface to determine if it shuts down or change state.
- Accepts all **activation notifications** from the activation manager and process accordingly.

Standby Server Functions

- Configures all internal data structures at startup time based on the ICCP configuration or domain data found in the ICCP ASR
- Receives all outgoing data change events (block 1 & 2) using the DDS server.
It does not send this data to the corresponding remote control center since there is no active connection.

- Receives all outgoing device control events (block 5) using the DDS server.
It does not send these control requests to the corresponding remote control center since there is no active connection.
- Listens on the ART callback interface to determine if it shuts down or change state.
- Accepts all **activation notifications** from the activation manager and process accordingly.
- If the ART/SSR informs the standby server to become the master server, it does the following:
 - Attempts to make an association with all remote control centers configured in the ICCP ASR.
 - Receives all incoming data from the remote control centers.
 - Uses the DDS Y-client interface to send all incoming data to the configured DDS server (most likely running on the real-time server).
 - Receives all outgoing data change events (block 1 & 2) using the DDS server.
It sends this data to the corresponding remote control center.
 - Receive all outgoing device control events (block 5) using the DDS server.
It sends these control requests to the corresponding remote control center.

The standby server does not:

- Attempt to make an association with any remote control centers configured in the ICCP ASR.
- Receive any incoming data from the remote control centers.

Failover Scenario

During a failover scenario (for example, ICCP master server failover), the scenarios are handled as follows:

- Data is not exchanged with the remote control center until the standby server has become master and has established communications with the remote control center
- After the data link connections have been reestablished, the normal ICCP protocol initialization is done to ensure that all data point values are synchronized. This initialize includes a full transfer of all data from each remote control center back to the Spectrum Power 5 ICCP node
- Any supervisory control actions that were pending at the time of the failover time-out and the operator or other application reissues these commands. The supervisory control software, recognizes the time-out and inform the applications appropriately
- There is no other synchronization done before or during a failover between the master and standby ICCP servers

The switchover time from the master server to the standby server takes a few seconds after being notified by ART. The ICCP processes on the standby server that is started and the configuration information read out of ICCP ASR. The subscription to DDS for data to send to the remote is completed.

The time it takes to reestablish the association and start exchanging data can vary from a few seconds to possibly minutes. This depends on several different factors as follows:

- A new association cannot be established until the remote control center detects that the previous association is broken.
- If the remote control center is the initiator, it delays before retrying to establish an association.
- Once the association is reestablished, it is necessary to renegotiate the transfer sets and then read the initial values. The amount of data and the speed of the communication medium have an impact on how long it takes.



NOTE

For additional information on failover schemes and general values like detection time and switchover time, refer to Technical Specification System Software (SYSW).

3.2.3 ICCP and EBBC

3.2.3.1 Overview

Spectrum Power 5 supports an Emergency Backup Control Center (EBCC) configuration consisting of a main control center and one or more emergency backup control centers.

For additional information, refer to Technical Specification System Software (SYSW).

3.2.3.2 ICCP Application

In an EBCC configuration, each of the control centers sites has server pairs running the ICCP software. On the active master control center, the ICCP software is actively creating or accepting connections and exchanging data.

For additional information on how ICCP behaves on active master control center, refer to [3.2.2 ICCP Server Redundancy](#).

On the control centers that are not active master control center (normally EBCC), the ICCP server pair has a master/standby redundancy designation. The **Master** ICCP server on the non-active control center functions like the **Standby** ICCP server on the active master control center. The **Master** ICCP server does not attempt to create ICCP connections, so it does not exchange data with the remote control centers.

For additional information, refer to [3.2.2 ICCP Server Redundancy](#).

When a control center becomes the active master control center, the **master** ICCP server on that control center reacts in the same way as how the **standby** server reacts on a server failover. Review the **Failover Scenario** in the previous section. It is realized that instead of failing over from the **master** ICCP server to the **standby** ICCP server within a control center, the failover can be from the **master** ICCP server in one control center to the **master** ICCP server on the new active master control center.

3.2.3.3 ICCP User Interface

The ICCP user interface shows the ICCP system, remote, and links administrative and operating state for the control center where the user interface is running. You can control the administrative states of the control center only when its active master control center.

3.2.4 Startup and Shutdown

Process Startup

When the ICCP server starts up, Spectrum Power 5 SSR startup the ICCP main process. It reads the system data and remote data. If the ICCP subsystem is enabled, it starts up an ICCP child process for each of the remotes configured for that ICCP Server. Each ICCP child process reads all the remote information associated with it. This includes the link data, the accessible data objects to set up the server function, and the transfer groups and transfer objects for the client function. When the ICCP child process starts, it starts up the **osill2d** process (and the snap process if security is configured).

Process Shutdown

When Spectrum Power 5 on the ICCP Server is shut down, the ICCP main and ICCP children processes also shuts down. It is also possible to shut down all the ICCP children by marking down the ICCP subsystem. The ICCP children stay down until the ICCP subsystem is enabled again.

It is also possible to stop individual ICCP children processes using the Spectrum Power 5 system management display. But the ICCP main detects the child process going away and restarts it.

Association Establishment

If the remote is enabled, the ICCP software establishes associations.

Transport Layer Security Authentication

If Transport Layer Security (TLS) ICCP security is configured for the link, the initiator sends a connection request to the remote systems secure TCP port, which is port 3782. The initiator receives and authenticates a

PKI certificate from the remote system. This authentication requires that a Certificate Authority (CA) certificate associated with the issuer of the remote systems certificate is known by the initiator. The CA certificate contains the public key that is used to validate the signature information in the received certificate. The information in the received certificate is also compared to the certificate information configured for this remote system.

If the initiator successfully authenticates the remote systems certificate, the initiator then sends its own PKI certificate to the remote system for authentication.

After the PKI certificate authentication is complete, the initiator and remote system agree on a cipher to use for data encryption. After this point all data on the connection is encrypted.

Application Authentication

If AR authentication (also called ACSE or MACE authentication) is configured for the association, then the client sends its Local AR certificate to the server for authentication. If the server successfully authenticates the client AR certificate, the server sends its own Local AR certificate to the client. If the TLS Authentication is configured, the AR certificate exchange for application authentication is encrypted.

ICCP Authentication

During the process of establishing of associations, the ICCP versions, bilateral table IDs, and the supported ICCP features are exchanged. The remote system minimally supports the conformance block 1, which provides association management and periodic data transmission functions.

Transfer Group Control

When configuring an association with the remote node, the data engineer configures measurements it wants to receive from the remote system in to transfer groups (also referred to as client block 1 & 2 containers). Measurements with the same reporting characteristics (like periodicity and buffer time) are placed in the same transfer group. ICCP maps these transfer groups to ICCP transfer sets and associated datasets. The ICCP dataset is a list of measurements and the transfer set is an ICCP structure that contains the information on how a dataset is reported. Because the definition of a transfer group can be large, ICCP requires to set up several datasets (and thus transfer sets) to hold all the measurements defined in a transfer group.

ART Interface

ICCP uses the Spectrum Power 5 ART interface to stop and start the ICCP process. The **Application Runtime (ART)** is an **Application Framework** that starts most of the ICCP components. Once started, all ICCP processes register with ART.

3.2.5 ICCP Security

ICCP security is based on IEC 62351-3 and IEC 62351-4 standards. The standards provide:

- Authentication of the client and server entities that are establishing an ICCP connection. The 2 types of authentication are:
 - Transport Layer Security (TLS) authentication
This uses **Public Key Infrastructure (PKI)** certificates that are exchanged by the client and server entities.
 - Application Layer authentication (also known as ACSE authentication or MACE authentication)
This provides authentication at the application layer. ACSE authentication also exchanges PKI certificates and matches application layer certificate information received from a remote system with configured certificate information.

- Encryption of the ICCP data exchanged between the local and remote systems. The cipher used to encrypt the data is negotiated between the client and server as part of the TLS protocol.
The bilateral agreement specifies, the types of authentication that are performed and the types of certificates that are used. ICCP allows the same certificate to be used for both TLS and ACSE authentication. It also supports the use of unique certificates for each type of authentication. ICCP security uses windows certificate store to manage the certificates. This ICCP security is supported only for ICCP associations using TCP/IP.

Authentication

ICCP security uses PKI certificates to authenticate associations and establish encryption.

Within the Spectrum Power 5 configuration of ICCP, certificates are categorized in the following terms:

- Local system certificates, which identify the local system and provide the basis for encrypted communication.
- Local and remote AR certificates, which is used to identify specific links (associations).
- Certificate authority certificates, which identify the creator of the other certificates.

Local system certificates and certificate authority certificates are required to implement ICCP security.

The Local system certificates are acquired from a trusted **Certificate Authority** (CA) and installed on their respective local systems. They are exchanged and verified to establish a secure association.

Local and remote AR certificates are negotiated as part of the bilateral agreement between 2 control centers. They, like the local system certificates, are acquired from a trusted CA. They are used for a second level of authentication and are used to determine the identity of the remote control center.

Certificate authority certificates are acquired from the certificate authorities that issued the local system, local AR, and remote AR certificates. It is possible for associating systems to get their certificates from different CA, so each system acquires enough CA certificates to verify the authenticity of each system (and Remote AR) certificates that it receives.

ICCP security uses windows certificate store to manage the certificates.

Configuration

The Spectrum Power 5 Information Model Management (IMM) configures ICCP security. The data engineer specifies, the security modes agreed upon in the bilateral agreement for each link. Certificate identification information is also entered into IMM so the certificate is retrieved from the windows certificate store by the ICCP communication stack. Spectrum Power 5 distributes the configuration parameters to each ICCP process requiring them.

ICCP communication stack accesses the certificates in the windows certificate store, as needed.

The customer performs management of the certificates within the windows certificate store.

3.2.6 DDS Events and Failure Conditions

After an initialization confirmation from ART, ICCP initializes with DDS. On the active ICCP server, ICCP registers as the producer for all ICCP block 1 and 2 data that are received from the remote ICCP node (client data). It also registers as the producer of the digital measurements that represent the operating states of the remote, link, and transfer groups.

ICCP subscribes for all ICCP block 1 and 2 data it provides to the remote (server data). It also subscribes for the digital measurements that represent the administrative states of the remote and transfer groups. It also subscribes as a customer for all block 5 control points.

3.2.7 Error Logging

The ICCP applications store their error logging information in to the event log.

4 Algorithms and Concepts

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4.1 ICCP Data Types

The following ICCP data types are supported in ICCP for exchange of block 1 & 2 data. The ICCP standard describes **Discrete** as a 32-bit integer and **Real** as a 32-bit floating point number with 8 of the bits being the exponent. The actual data type names change depending upon Spectrum Power 5 naming convention but the description is the same (in IMM, the term **Analog** is used instead of **Real**).

Table 4-1 ICCP Data Types

ICCP Data Types	Types Description
Real RealQ RealQTimeStamp RealQTimeampExtended RealExtended	Represents a real value without quality or time stamp Represents a real value with quality Represents a real value with quality and time stamp (resolution to the second) Represents a real value with quality and a time stamp that includes millisecond resolution Represents a real value with quality, time stamp with resolution to the second, and Change of Value (COV) Counter
Discrete DiscreteQ DiscreteQTimeStamp DiscreteQTimeStampExtended DiscreteExtended	Represents a discrete value without quality or time stamp Represents a discrete value with quality Represents a discrete value with quality and time stamp (resolution to the second) Represents a discrete value with quality and a time stamp that includes millisecond resolution Represents a discrete value with quality, time stamp with resolution to the second, and COV Counter
State StateQ StateQTimeStamp StateQTimeStampExtended StateExtended	Represents a state value without quality or time stamp Represents a state value with quality Represents a state value with quality and time stamp (resolution to the second) Represents a state value with quality and a time stamp that includes millisecond resolution Represents a state value with quality, time stamp with resolution to the second, and COV Counter

Type conversions occur on the data sent and received across ICCP. Data received across the ICCP is converted to the proper Spectrum Power 5 SCADA type, and then encoded before uploaded to DDS. Data received from DDS is decoded and then converted to the configured ICCP data type.

The ICCP real and discrete data types can be used with most of the Spectrum Power 5 SCADA types, but realize the consequences.

For example, the discrete data type is used for sending **Analog Data**, but anything after the decimal point is lost.

Table 4-2 Most Commonly Used ICCP Data Types

Spectrum Power 5 SCADA Type	ICCP Data Type
Analog Data	Real RealQ RealQTimeStamp RealQTimeStampExtended
Discrete Data	State StateQ StateQTimeStamp StateQTimeStampExtended Discrete DiscreteQ DiscreteQTimeStamp DiscreteQTimeStampExtended
Accumulator Data	Discrete DiscreteQ DiscreteQTimeStamp DiscreteQTimeStampExtended Real RealQ RealQTimeStamp RealQTimeStampExtended
Discrete Data (Tap Position)	Discrete DiscreteQ DiscreteQTimeStamp DiscreteQTimeStampExtended

4.2 Interoperability List

4.2.1 Basic Protocol and Services Conformance

This section describes various ICCP features directly related to Section 9 of IEC 60870-6-503 Ed. 2.0 TASE.2 Services and Protocol, Version 2000-08.



NOTE

Each row includes an indication whether the feature is mandatory (m) or optional (o). Sometimes the indication is supplemented by a number that indicates the associated conformance block. If 2 m / o indications separated by comma are provided, the left indication applies to operation in client mode and the right indication to operation in server mode.



NOTE

The **ICCP Basic Code** is a set of libraries on which the Spectrum Power 5 ICCP product is built. The columns under **ICCP Basic Code** and **Spectrum Power 5 ICCP** indicate what layer of the product supports (Y) or does not support (N) the feature when operating either in client or server modes.

You can see some rare cases in which a feature is supported in the ICCP Basic Code column but not in the Spectrum Power 5 ICCP column. Spectrum Power 5 standard ICCP product conforms to what is listed under the Spectrum Power 5 ICCP columns.

Product Identification

Vendor Name	SIEMENS
Product Name	TASE.2
Model	NT
Revision	3.0
Other Identification	
Client Supported	Y
Server Supported	Y

Conformance Building Block

The following is a tabular summary of section 9.1 Allowable Subsets™ of the IEC 60870-6-503 publication.

			ICCP Basic Code		Spectrum Power 5 ICCP	
	Conformance Block Number	Mandatory/ Optional	Client	Server	Client	Server
Basic Services	1	m	Y	Y	Y	Y
Extended Conditions	2	o	Y	Y	Y	Y
Blocked Transfers	3	o	Y	N	N	N
Information Message	4	o	Y	N	N	N
Device Control	5	o	Y	Y	Y	Y
Programs	6	o	N	N	N	N
Events	7	o	N	N	N	N
Accounts	8	o	Y	Y	N	N
Time Series	9	o	N	N	N	N

PICS

The following tables correspond to the tables in section 9.2 PICS™ of the IEC 60870-6-503 publication.

Association Management

		ICCP Basic Code		Spectrum Power 5 ICCP	
	Mandatory/ Optional	Client	Server	Client	Server
Supported Features	m1	Y	Y	Y	Y
QOS	o	Y	Y	N	N
Associate Operation	m1	Y	Y	Y	Y
Conclude Operation	m1	Y	Y	Y	Y
Abort Operation	m1	Y	Y	Y	Y

Data Values

		ICCP Basic Code		Spectrum Power 5 ICCP	
	Mandatory/ Optional	Client	Server	Client	Server
Data Value Model	m1	Y	Y	Y	Y
VCC-specific scope*	m1, o	Y	Y	Y	Y
ICC-specific scope *	m1, o	Y	Y	Y	Y
Get Data Value Oper- ation	o, m1	Y	Y	Y	Y
Set Data Value Oper- ation	o, m1	Y	Y	N	N
Get Data Value Names Operation	o, m1	Y	Y	Y	Y
Get Data Value Type Operation	o, m1	Y	Y	Y	Y

* Either the VCC-specific or ICC-specific scope is supported by servers for block 1.

Datasets

		ICCP Basic Code		Spectrum Power 5 ICCP	
	Mandatory/ Optional	Client	Server	Client	Server
Data Set Model	m1	Y	Y	Y	Y
VCC-specific scope*	m1, o	Y	Y	N	Y
ICC-specific scope*	m1, o	Y	Y	N	Y
Create Data Set Opera- tion	o	Y	Y	Y	Y
Delete Data Set Opera- tion	o	Y	Y	Y	Y
Get Data Set Element Values Operation	o, m1	Y	Y	Y	Y
Set Data Set Element Values Operation	o, m1	Y	Y	N	N
Get Data Set Names Operation	o, m1	Y	Y	Y	Y
Get Data Set Element Names Operation	o, m1	Y	Y	Y	Y

*Either VCC-specific or ICC-specific scope is supported by servers for block 1.

Accounts

		ICCP Basic Code		Spectrum Power 5 ICCP	
	Mandatory/ Optional	Client	Server	Client	Server
Account Model	m8	Y	Y	N	N
Query Operation	m8	N	Y	N	N

DS Transfer Sets

		ICCP Basic Code		Spectrum Power 5 ICCP	
	Mandatory/ Optional	Client	Server	Client	Server
Data Set Transfer Set Model	m1	Y	Y	Y	Y
Start Transfer Operation	m1	Y	Y	Y	Y
Stop Transfer Operation	m1	Y	Y	Y	Y
Get Next DSTransfer Set Operation	m1	Y	Y	Y	Y
IntervalTimeOut	o, m1	Y	Y	Y	Y
ObjectChange	o, m2	Y	Y	Y	Y
OperatorRequest *	o, m1	N	Y	N	N
IntegrityTimeout	o, m2	Y	Y	Y	Y
OtherExternalEvent *	o, m1	N	N	N	N
EventCodeRequested *	o, m1	N	N	N	N
StartTime	m1	Y	Y	Y	Y
Interval	m1	Y	Y	Y	Y
TLE	o, m2	N	N	N	N
BufferTime	o, m2	Y	Y	Y	Y
Integrity Check	o, m2	Y	Y	Y	Y
DSConditions Requested	o, m1	Y	Y	Y	Y
Block Data	o, m3	N	N	N	N
Critical	o, m2	Y	Y	Y	Y
RBE	o, m2	Y	Y	Y	Y

* Servers must support the processing of these parameters. This does not imply the presence of the application.

Time Series Transfer Set Objects

		ICCP Basic Code		Spectrum Power 5 ICCP	
	Mandatory/ Optional	Client	Server	Client	Server
Time Series Transfer Set Model	m9	N	N	N	N
Get Next TSTransfer Set Operation	m9	N	N	N	N
EndTimeArrived	o, m9	N	N	N	N

		ICCP Basic Code		Spectrum Power 5 ICCP	
	Mandatory/ Optional	Client	Server	Client	Server
ReportIntervalTimeOut	o, m9	N	N	N	N
OperatorRequest	o, m9	N	N	N	N

Transfer Account Transfer Set Objects

		ICCP Basic Code		Spectrum Power 5 ICCP	
	Mandatory/ Optional	Client	Server	Client	Server
Transfer Account Transfer Set Model	m8	Y	Y	N	N
BeforeTheHour	o, m8	Y	Y	N	N
DispatchUpdate	o, m8	Y	Y	N	N
DuringTheHour	o, m8	Y	Y	N	N
AfterTheHour	o, m8	Y	Y	N	N
ActualDataUpdate	o, m8	Y	Y	N	N
PastHours	o, m8	Y	Y	N	N
ObjectChange	o, m8	Y	Y	N	N
OperatorRequest	o, m8	Y	Y	N	N

Information Message Objects

		ICCP Basic Code		Spectrum Power 5 ICCP	
	Mandatory/ Optional	Client	Server	Client	Server
Information Message Transfer Set Model	m4	Y	Y	N	N

Special Transfer Set Objects

		ICCP Basic Code		Spectrum Power 5 ICCP	
	Mandatory/ Optional	Client	Server	Client	Server
Transfer Set Name	o, m1	Y	Y	Y	Y
Next DSTransfer Set	o	Y	Y	N	N
Next TSTransfer Set	o	N	N	N	N
Event Code	o, m1	N	Y	N	Y
DSConditionsDetected	o, m1	Y	Y	Y	Y
TSTransferSetDetected	o, m1	N	N	N	N
TACConditionsDetected	o, m1	Y	Y	N	N
Transfer Set Time Stamp	o, m1	Y	Y	N	N

Devices

Table 4-3 Devices

		ICCP Basic Code		Spectrum Power 5 ICCP	
		Client	Server	Client	Server
Device model	m5	Y	Y	Y	Y
Select Operation	m5	Y	Y	N	N
Operate Operation	m5	Y	Y	Y	Y
Get Tag	o	Y	Y	N	N
Set Tag	o	Y	Y	Y	Y
Time-out Action	o, m7	N	N	N	N
Local Reset Action	o	N	N	N	N
Success Action	o, m7	N	N	N	N
Failure Action	o, m7	N	N	N	N

Programs

		ICCP Basic Code		Spectrum Power 5 ICCP	
		Client	Server	Client	Server
Program Model	m6	N	N	N	N
Start Operation	m6	N	N	N	N
Stop Operation	m6	N	N	N	N
Resume Operation	m6	N	N	N	N
Reset Operation	m6	N	N	N	N
Reset Operation	m6	N	N	N	N
Kill Operation	m6	N	N	N	N
Get Program Attributes Operation	o, m6	N	N	N	N

Event Enrollments

		ICCP Basic Code		Spectrum Power 5 ICCP	
		Client	Server	Client	Server
Event Enrollment Model	m7	N	N	N	N
Create Event Enrollment Operation	m7	N	N	N	N
Delete Event Enrollment Operation	o	N	N	N	N
Get Event Enrollment Attribute Operation	o	N	N	N	N

Event Conditions

		ICCP Basic Code		Spectrum Power 5 ICCP	
	Mandatory/ Optional	Client	Server	Client	Server
Event Condition Model	m7	N	N	N	N
Event Notification Action	m7	N	N	N	N

4.2.2 Objects Models Conformance

This section describes various ICCP features directly related to Section 9 in IEC 60870-6-802 Ed. 2.0 TASE.2 Object Models, Version 2000-08.

The following tables correspond to the tables in Section 9 Conformance™ of the IEC 60870-6-802 publication.



NOTE

Each row includes an indication whether the feature is mandatory (m) or optional (o). Sometimes the m / o indication is supplemented by a number that indicates the associated conformance block. If 2 m / o indications separated by comma are provided, the left indication applies to operation in client mode and the right indication to operation in server mode.



NOTE

The **ICCP Basic Code** is a set of libraries on which the Spectrum Power 5 ICCP product is built. The columns under **ICCP Basic Code** and Spectrum Power 5 ICCP indicate what layer of our product supports (Y) or does not support (N) the feature when operating either in client or server modes. You can see some rare cases in which a feature is supported in the **ICCP Basic Code** column but not in the Spectrum Power 5 ICCP column. To avoid confusion, focus on the Spectrum Power 5 ICCP columns.

Supervisory Control and Data Acquisition

		ICCP Basic Code		Spectrum Power 5 ICCP	
	Mandatory/ Optional	Client	Server-	Client	Server
Indication Point Object	m1	Y	Y	Y	Y
Control Point Object	m5	Y	Y	Y	Y
Protection Equipment Event Object	o	N	N	N	N

Transfer Accounts

		ICCP Basic Code		Spectrum Power 5 ICCP	
	Mandatory/ Optional	Client	Server Client	Client	Server Client
TransferAccount Object	m8	Y	Y	N	N
TransmissionSegment Object	m8	N	N	N	N
ProfileValue Object	m8	Y	Y	N	N
AccountRequest Object	o	Y	Y	N	N

Device Outage

		ICCP Basic Code		Spectrum Power 5 ICCP	
	Mandatory/ Optional	Client	Server	Client	Server
Device Outage Object	o	Y	Y	N	N

Information Message

		ICCP Basic Code		Spectrum Power 5 ICCP	
	Mandatory/ Optional	Client	Server	Client	Server
Information Message	m4	Y	Y	N	N

Power Plant

		ICCP Basic Code		Spectrum Power 5 ICCP	
	Mandatory/ Optional	Client	Server	Client	Server
Availability Report Object	o	N	N	N	N
RealTime Status Object	o	N	N	N	N
ForecastSchedule Object	o	N	N	N	N
Curve Object	o	N	N	N	N
Power System Dynamic Object	o	N	N	N	N

GeneralDataReport

		ICCP Basic Code		Spectrum Power 5 ICCP	
	Mandatory/ Optional	Client	Server	Client	Server
GeneralDataReport Object	o	N	N	N	N
GeneralDataResponse Object	o	N	N	N	N

4.3 ICCP Conformance Blocks

4.3.1 Block 1

Object	Services	Supported by Spectrum Power 5 Server	Supported by Spectrum Power 5 Client*
Association		yes	yes
	Initiate	yes	yes
	Conclude	yes	yes
	Abort	yes	yes
Data Value		yes	yes
	Get Data Value	yes	yes
	Set Data Value	no	no
	Get Data Value Names	yes	yes
	Get Data Value Type	yes	yes
Data Set		yes	yes
	Create Data Set	yes	yes
	Delete Data Set	yes	yes
	Get Data Set Element Values	yes	yes
Services	Set Data Set Element Values	no	no
	Get Data Set Names	yes	yes
	Get Data Set Element Names	yes	yes
DSTransfer Set		yes	yes
	Start Transfer	yes	yes
	Stop Transfer	yes	yes
	Condition Monitoring - IntervalTimeOut	yes	yes
	Condition Monitoring - Operator Conditions **	no	no*
Next Transfer Set		yes	yes
	Get Next Transfer Set Value	yes	yes

* A no is not an indication of non-conformance with the ICCP standard. Instead, it indicates only that the ICCP client does not use this service, typically because the service is normally not performed by a client.

** The ICCP server supports the required protocol elements, but does not supply a user interface for the server operator.

4.3.2 Block 2

The client and server functions for Block 2 are the same as those for Block 1 except the condition monitoring of the **ObjectChange**.

For additional information, refer to Section [2.2 Block 2 - Conditions](#).

4.3.3 Block 5

Object	Services	Supported by Spectrum Power 5 Server	Used by Spectrum Power 5 Client
Device			
	Select	no	no
	Operate	yes	yes
	Set Tag	yes	yes
	Get Tag	no	no

4.4 Security Certificate Management

Within the Spectrum Power 5 configuration of ICCP, certificates are categorized in the following terms:

- Local System Certificates, which identify the local system and provide the basis for encrypted communication.
- Local and Remote AR Certificates, which can be used identify specific links (associations).
- Certificate Authority Certificates, which identify the creator of the other certificates.

The certificates are installed in a Windows certificate store using the Certificates MMC (Microsoft Management Console) snap-in.

Local System Certificates and local AR Certificates are imported into MMC from .pfx files (Personal Information Exchange-PKCS #12). These .pfx files contain the certificate and its associated private key. These files require a private key password.

Certificate Authority (CA) Certificates are imported into MMC from .cer files. These files contain the CA identification and public key to validate received certificates that were issued by this CA.

The loading and management of the certificates in the Microsoft Certificate Store is the responsibility of the customer.

4.5 Supervisory Control

Supervisory Control Data Mapping

ICCP block 5 device objects consist of one of the following value types:

- Command
- Set Point
 - Set Point Real
 - Set Point Discrete

Normally, the ICCP command is associated with a Spectrum Power 5 **Discrete (non Tap Position)** value. The ICCP **Set Point Real** associated with an **Analog Measurement**. The ICCP **Set Point Discrete** associated with **Discrete (Tap Position) Measurement**.

Supervisory Control Step Position

ICCP maps the Spectrum Power 5 Step Position Control to an ICCP Command. The **STEP UP** command is mapped to the ICCP command **Raise**. The **STEP DOWN** command is mapped to the ICCP command **Lower**.

4.6 Applicable Standards

- **IEC 60870-6-503, Ed. 2.0: Telecontrol equipment and systems - Part 6-503: Telecontrol protocols compatible with ISO standards and ITU-T recommendations - TASE.2 Services and protocol**
- ISO/IEC 60870-6-503, TASE.2 Services, and Protocol, Version 1996-08, ICCP, Inter-Control-Center Communications Protocol (ICCP)
- Inter-Control Center Communications Protocol (ICCP), Section 503: Service and Protocol Version 6.1, August 1996
- Inter-Control Center Communications Protocol (ICCP), Section 702: Functional Profile Version 5.1, November 1994
- **IEC 60870-6-802, Ed. 2.0: Telecontrol equipment and systems - Part 6-802: Telecontrol protocols compatible with ISO standards and ITU-T recommendations - TASE.2 Object models**
- Inter-Control Center Communications Protocol (ICCP), Section 802: Object Models Version 6.1, August 1996
- ISO/IEC 60870-6-802, TASE.2 Object Models, Version 1996-08, ICCP, Inter-Control Center Communications Protocol (ICCP)
- ISO/IEC 9506-1 1990, Industrial Automation Systems - Manufacturing Message Specification Part 1: Service definition
- ISO/IEC 9506-2 1990, Industrial Automation Systems - Manufacturing Message Specification Part 2: Protocol Specification

4.7 Spectrum Power 5 Interpretation of the TASE.2 Tags Definition

This section presents Spectrum Power 5 interpretation and implementation of the TASE.2 definition of tags. The TASE.2 User Guide (60870-6-505) states that the Tag Value **indicates whether a Control Point is tagged, and if it is, what the level of tagging is**. The Spectrum Power 5 implementation attempts to expand this definition to provide more functionality.

It is important that Spectrum Power 5 users who want to exchange tags with non-Spectrum Power 5 implementations be aware of this implementation, so they can determine, if there is interoperability issues.

The TASE.2 Definition of Tags:

In the 2000-8 revision of TASE.2, the ability to report tags via Block 1&2 was added. Previously tags were only a Block 5 action. For both Block 5 and Block 1&2, the same simple structure is used, which defines only 3 possible Tag Values (one value being NO-TAG).

The IEC 60870-6-802 document defines a Tag_Value structure that consists of 2 COMPONENTS (fields):

- **Flags**
The Flags field is defined as a bit string, consisting of a Tag_hi (bit 0), Tag_lo (bit 1), and Tag_state (bit 2). The Tag_hi and Tag_lo combined hold the value of 0=NO-TAG, 1=OPEN-AND-CLOSE-INHIBIT, or 2=CLOSE-ONLY-INHIBIT. The Tag_state bit has the value of 0 = IDLE or 1 = ARMED.
- **Reason**
The **Reason** text box is defined as a VisibleString {width 255}. A character in a VisibleString must be in the range of 31 < valid char < 127.

The definition is best suited for communicating the level of tagging on a point, which means a **Tag Value** can have only one value for a point, either NO-TAG, Open-And-Close-Inhibit, or Close-Only-Inhibit. It was not defined to report the existence of both an **Open-And-Close-Inhibit** tag **AND** a **Close-Only-Inhibit** tag on a point. The definition does not indicate that 3 is a valid value (implying both exist). Even if it did allow for both bits to be set, there is only 1 **Reason** field, so the standard needs to define which tag the text in the **Reason** field belong to.

Spectrum Power 5 Interpretation and Implementation:

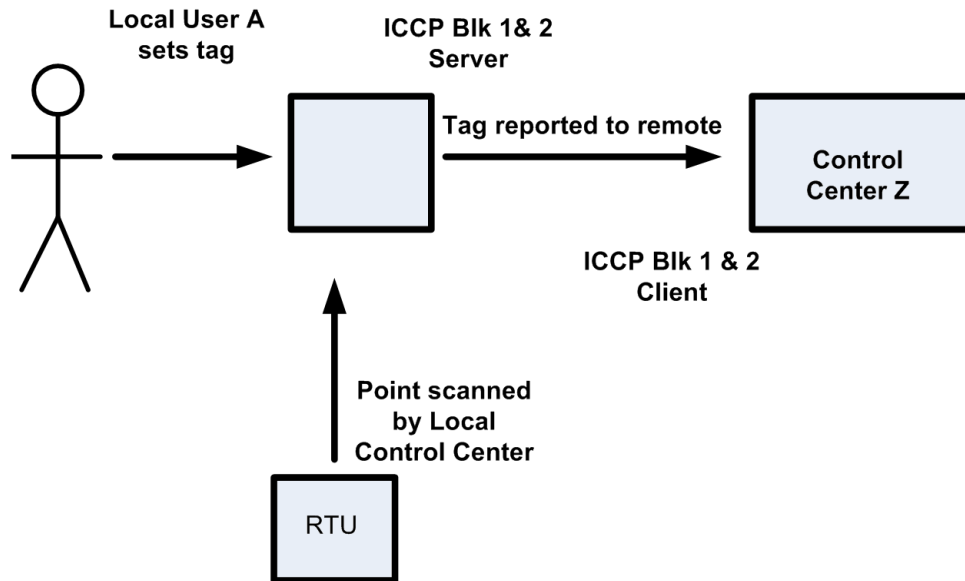
The Spectrum Power 5 system allows you to set multiple tags of different types on a point.

For example, one operator can set a **Control Inhibit** tag on a point, and a different operator can set a **Control Inhibit Close** tag on the same point, so both exist at the same time. (SIEMENS normally equates the **TASE.2 Open-And-Close-Inhibit** to the Spectrum Power 5 Control Inhibit and the **Close-Only-Inhibit** to the Spectrum Power 5 Control Inhibit Close).

The Spectrum Power 5 system attempts to use the TASE.2 definition to propagate the existence of both tag types to the other system. It attempts to do this without violating the TASE.2 definition of just reflecting the tag level. In order to do this, Spectrum Power 5 presents these clarifications on its interpretation of the standard.

Tags Reported Via Block 1 and 2 Versus Block 5:

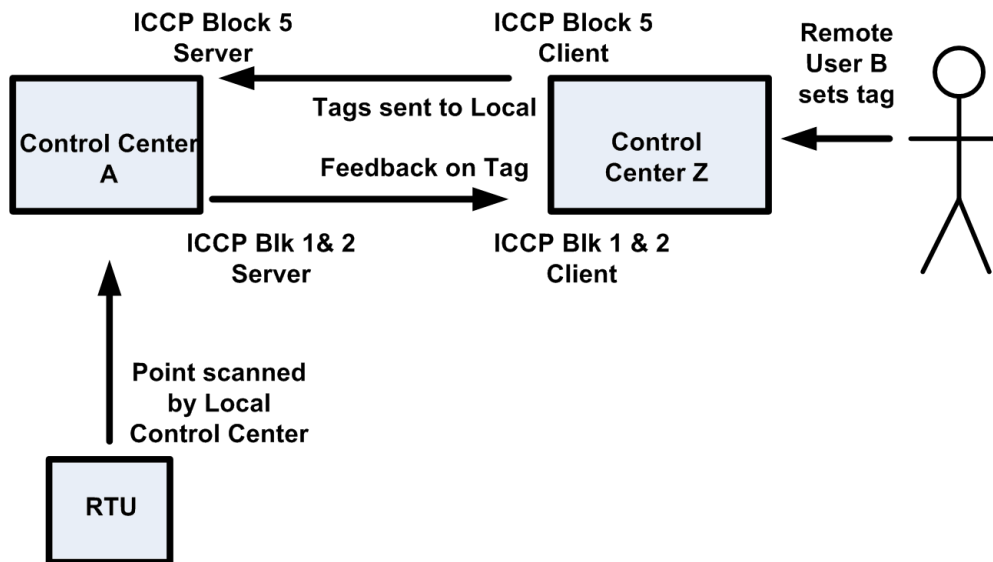
Block 1 and 2 is used to report when a tag is set on the system that directly controls the device. In the following figure, the user on **Control Center A** places a tag on a point that it provides to **Control Center Z** (Control Center A **Owns** point). The **Tag** information is reported to **Control Center Z** using Block 1 and 2 reporting. In this example, Block 5 is NOT used to send a tag from **Control Center A** to **Control Center Z**.



[tsiccp04-050309-02.tif, 1, en_US]

Figure 4-1 Tags for Block 1 and 2

Block 5 is used when the control center that does NOT directly control the devices wants to set a tag on the Control Center that does control the device. In the following figure, you are on a **Control Center Z** places a Tag on a point that is provided by **Control Center A** (**Control Center A Owns** point). A Block 5 Set Tag control request is used to set the Tag on **Control Center A**. When Block 5 is used to set a tag, it is recommended that Block 1 and 2 also be configured to provide feedback that the tag was properly set.



[tsiccp05-050309-02.tif, 1, en_US]

Figure 4-2 Tags for Block 5

Tag Value Levels

The standard indicates it is only reporting the tag level on a point; it is not clear which tag level is the highest. It would seem that the **Open-And-Close-Inhibit** is a higher level than the **Close-Only-Inhibit**. But based on the actual value, it could be argued the **Close-Only-Inhibit** is higher (NO-TAG = 0, Open-And-Close-Inhibit = 1, and Close-Only-Inhibit = 2).



NOTE

Spectrum Power 5 assumes **Open-And-Close-Inhibit** is the highest tag level.

Based on the TASE.2 definition, if an **Open-And-Close-Inhibit** tag is set on a point, that is the value that gets transmitted. If a **Close-Only-Inhibit** is then set on a point, the standard does not require this indication be transmitted, because it is a lower-level tag.

If the **Spectrum Power 5** server receives a read request for a tag point, it returns the Tag Level. If both tags are set on the server, Spectrum Power 5 can only return the **Open-And-Close-Inhibit** indication (the higher tag level). The same is true if the server provides a periodic report of the value (such as an integrity report for Block 1&2 reporting). The server can only report one value, the Tag Level.

Reporting the Existence of Both Types of Tags

The 2000-08 revision of the TASE.2 implements the AllChangesReported capability. Spectrum Power 5 uses this capability in its Block 1&2 reporting of Tags to report more than just the Tag Level. It attempts to report the existence of both types of tags. If one operator sets an **Open-Close-Inhibit**, and another operator immediately sets a **Close-Only-Inhibit** tag on the same point, the Spectrum Power 5 server reports 2 Tag Values, one containing the Open-Close-Inhibit and the other the **Close-Only-Inhibit**.

The same is true for Spectrum Power 5 client when you attempt to set a **Tag** using Block 5. When you set an **Open-Close-Inhibit** tag, Spectrum Power 5 performs a **Set Tag** operation with the **Open-Close-Inhibit** for the **Tag Value**. If a **Close-Only-Inhibit** is set on the same point, the Spectrum Power 5 client performs another **Set Tag** operation with the **Close-Only-Inhibit** value.

The State Field in Flags That Indicate a Tag Should be Set/Cleared

An issue with attempting to report the existence of both types of tags versus just reporting the tag level is how to report a tag has been cleared. If the server only supports the reporting the tag level, if the **Open-Close-Inhibit** tag is cleared, it just reports NO-TAG, or if a **Close-Only-Inhibit** tag exists, it reports the **Close-Only-Inhibit** tag level. By just reporting the lower level, it implies the higher level tag has been cleared. There is no need for a clear action.

But because Spectrum Power 5 is attempting to report the existence of both types of tags, it reports a particular type has been cleared. To do this, Spectrum Power 5 uses the tag_state field in the TASE.2 Tag_Value structure. The TASE.2 standard defines the tag_state field as having the meaning of Idle and Armed. These terms seem to imply the state of the device, and would not apply to tags. But because its part of the Tag Value structure, and because the SBO structure has a **Select State** indication to reflect the select state, Spectrum Power 5 uses the state field for indicating a tag has to be set or cleared.

When a **Close-Only-Inhibit** tag is set on the server, Spectrum Power 5 reports using Block 1 and 2 the Tag Value with the tag_state field set to Armed. If the **Close-Only-Inhibit** is deleted on the server, and an **Open-Close-Inhibit** tag still exists, Spectrum Power 5 reports the Tag Value of **Close-Only-Inhibit** with the tag_state set to Idle.

The Spectrum Power 5 client looks at the State field to determine, if the Tag Value should be created or deleted. If the client receives a NO-TAG for a point via Block 1 and 2, regardless of the State field, it clears all tags received from this remote that exist on the point. If the value is **Open-Close-Inhibit** or **Close-Only-Inhibit**, the client looks at the value of the State field to determine if this is a set or clear action.

Keeping the Systems in Sync (Block 1 and 2)

If only the tag level is being propagated, it is easy to keep the 2 systems in synchrocheck. If the server had been reporting the **Open-Close-Inhibit** tag level, and now reports the **Close-Only-Inhibit** tag level, this implies the Open-Close-Inhibit tag level has been cleared. So a single periodic report or a read response indicates to the client the tag level.

With the Spectrum Power 5 solution of Tags in Block 1 and 2, it attempts to report the existence of different types of tags. Thus, when Spectrum Power 5 reports the **Close-Only-Inhibit** with the tag_state flag indicating Armed, the **Open-Close-Inhibit** may still exist. So the client cannot assume the **Open-Close-Inhibit** tag has been cleared.

This is where the **AllChangesReported** functionality described earlier becomes important. If the **Close-Only-Inhibit** tag is cleared on the server, it sends out the **Close-Only-Inhibit** tag with tag_state set to Idle. If the **Open-Close-Inhibit** tag still exists, the server also sends out a value with **Open-Close-Inhibit** tag and tag_state set to Armed. This can only work with **AllChangesReported** functionality enabled, otherwise only one of the values is transmitted. If a client wants to receive Tag indications from the server, it must set up the transfer group to request **AllChangesReported** from the server.

When the connection with the remote is established, the client normally reads the current values. For Block 1 and 2, tags are just another value. The read of the Tag returns the Tag level for the point. But after this, if the **AllChangesReported** is set, the ICCP server will generate a report the initial state of both tag types.



NOTE

While the Spectrum Power 5 ICCP implementation attempts to report the existence of different tag types on a single point, this indication only reflects the existence of the tag type, not multiple instances of the same type on a single point.

In Spectrum Power 5, not only can the operator set different types of tags on a point, they can create multiple instances of the same type on a point. For example, an operator can create multiple Control Inhibits tags on a point. Spectrum Power 5 ICCP does NOT attempt to propagate each of these Control Inhibit instances to the other system. If multiple Control Inhibit instances do exist on a point, ICCP propagates the **Open-And-Close-Inhibit Tag Value**, and for the reason text, it uses the text from the oldest Control Inhibit tag instance on the point. As long as a Control Inhibit tag exists on the point, the server reports the **Open-And-Close-Inhibit Tag Value**.

To summarize, the Spectrum Power 5 solution attempts to keep the existence of tag types in sync with the remote system. It does NOT attempt to keep multiple instances of the same tag type on a point in sync with the remote.

Keeping the Systems in Sync (Block 5)

Keeping tags set using Block 5 is difficult. For block 5, the Spectrum Power 5 client only sends a request in response to an operator performing an action. If the operator sets the first **Open-And-Close-Inhibit** tag on a point, the Spectrum Power 5 client sends the **Tag Value** with the **Open-And-Close-Inhibit** value (and tag_state set to Armed). When the last instance of the **Open-And-Close-Inhibit** tags on a point has been removed, the client sends a clear indication. There is no periodic report or **AllChangesReported** aspect of the Block 5 tag to help ensure the remote is in synchrocheck

One thing that occurs with Block 5 that does not occur with Block 1 and 2 reporting is that there is immediate feed back on the Set Tag operation, that indicates, if the tag has been successfully set or not. If ICCP client is unable to set a tag on the remote system, it generates an alarm.

The Spectrum Power 5 client does do in an attempt to get the tags in sync occurs every time the ICCP association with the remote is reestablished. The client fires the tags types that currently exist.

Reason Field and Source of Text

The ICCP standard definition for the Reason field is VisibleString {width 255}. The Spectrum Power 5 implementation places the **Title** field from the **New Note Dialog** into the **Reason** text. If no note is created for the tag, or the **Title** is left blank, no **Reason** text is included with the tag.

The **Reason** text is only 255 characters long. If the **Title** you entered is longer than 255 characters, the text is truncated.

If **Reason** text is received with a tag, it is used to create a note and uses the **Reason** text for the **Title**. If no **Reason** text is provided, no note is created for the tag.

Reason Field and Non Latin Characters

Spectrum Power 5 has customers who create a tag note that contains characters in the **Title** field that do not fall within the range defined by the ICCP standard (a specific case is a customer using Cyrillic characters). The standard only allows characters in this range: $31 < \text{valid char} < 127$.

The Spectrum Power 5 solution for handling **Reason** character strings that contains characters outside the range supported by ICCP is to perform 2 encodings. The first encoding (mapping) is to go from UTF-16 to the Cyrillic (Windows 1251) character set. This reduces the characters from 2 bytes down to 1 byte (8 bits). Next, Base 64 encoding is applied so the data fits into 7 bits. This expands the data a little (so fewer characters can be transmitted).

To ensure that the receiving system is aware of this encoding, the Spectrum Power 5 implementation puts an identifier at the beginning of the **Reason** character string. The identifier has this format and meaning:

SIX<title>Title Text from Tag

SI – These are fixed characters, **SI**. They indicate this as a **Tag** from a Spectrum Power 5 system and that the **Reason** field has additional meaning. (SI is short for Spectrum Power 5 ICCP).

X – This identifies the encoding of the remaining text portion of the message (<title>will NOT be encoded). For now, X represents the characters 1 – 9. For customers who creates tags with characters outside of the ICCP range for Visible text, (Cyrillic characters, for example), ICCP requires to encode the text. This value X indicates which character set (code page) was used to encode the remaining text.

Here are the settings that are supported initially for X:

- 1 – Cyrillic (Windows 1251) character set.
ICCP does 2 encodings. The first encoding (mapping) is to go from UTF-16 to the Cyrillic (Windows 1251) character set. This reduces the characters from 2 bytes down to 1 byte (8 bits). Next, base-64 encoding is applied so the data fits into 7 bits. This expands the data a little (so fewer characters can be transmitted).

The receiving side looks at the X setting to determine what decoding, is done. If it is a 1, the base-64 decoding is done first, followed by the Cyrillic to UTF-16.

<title> The string following this keyword, up to the end of the message, is from the Spectrum Power 5 **Title** field that you set in the **Note** dialog. This is from the oldest tag instance of the specific type on the specific point. The string following <title> must be encoded as described. The ICCP standard only allows the **Reason** field to be 255 characters long. So it is possible that the **Title** you have entered is truncated when placed into the ICCP **Reason** field.

The need to perform the mentioned encoding is a configurable option defined at the remote level. It applies to all **Reason** character strings exchanged with the indicated remote. The default is no encoding so the prefix is not included in the reason field.

ICCP will receive/retrieve the text message from SCADA as UTF-16. If the ICCP remote is NOT configured to perform special processing on the text, ICCP truncates the high-order byte of the UTF-16 character. ICCP goes through the string making sure that all characters are in the $31 < \text{char} < 127$ range. Any characters that are not in this range is converted to a blank (decimal 32).

Mapping of Spectrum Power 5 Tag to TASE.2 Tag

As previously stated, the ICCP Standard only defines 2 tag values (3 if you count NO-TAG). The 2 tag values defined for TagFlags are **OPEN-AND-CLOSE-INHIBIT** and **CLOSE-ONLY-INHIBIT**. Spectrum Power 5 supports tags like these (Control Inhibit and Control Inhibit Close) and many more tags and markers. The Spectrum Power 5 implementation allows the customer to map different tags and markers to the 2 definitions defined by the ICCP Standard for TagFlags.

There is nothing in the message that indicates the different definition of the TagFlags value. The 2 systems exchanging the TagFlags with different meanings has to come to an agreement on how to map these bits. How this agreement is made is outside the scope of this implementation.

4.8 Mapping of SCADA Quality Flags to ICCP Quality Flags

ICCP Quality mapping (Scada to ICCP)											
Scada Quality Flags			ICCP Quality Flag								
Bit	Quality Flag	Meaning	Validity				Source				Normal Value
			VALID	HELD	SUSPECT	NOT VALID	TELEMETERED	CALCULATED	ENTERED	ESTIMATED	NORMAL / ABNORMAL
0	Telemetry blocked local	yes/ no		>0							
1	Telemetry blocked remote	yes/ no		>0							
2	Not Updated	yes/ no			>0						
3	Invalid local	yes/ no				>0					
4	Invalid remote	yes/ no				>0					
5	Suspect local	yes/ no			>0						
6	Suspect remote	yes/ no			>0						
7	Selection Manually Adjusted	yes/ no									
8, 9	Source Selection	00: Primary 01: Secondary 10: Tertiary 11: Auxiliary									
10, 11	Value Source	00: Telemetered 01: Calculated 10: Estimated 11: Entered					1	1		1	
12, 13	Value Source Detail	Value Source = Telemetered 00: Normal 01: Spontaneous 10: General Interrogation 11: Remote Control Response Value Source = Calculated 00: Operands valid 01: Operand(s) manually entered 10: Operand(s) suspect 11: Operand(s) invalid Value Source = Estimated Unobservable yes/ no Bad data yes/ no Value Source = Entered 0: Local 1: Remote							1	1	
14	True Time	yes/ no									
15	Object in test	yes/ no		>0							
16	Clapped to Zero	yes/ no			>0						
17	Initial Data	yes/ no									
18	Not in normal state	yes/ no									-1 1
19	Calculated by RAP auth. processing	yes/ no									
20, 21	RAP authorization source	00: Main 01: Control 10: Integral 1 11: Integral 2									
22	Incomplete aggregation	yes/ no									
23	Alarm suppress	yes/ no									
24	Free bits for future use										
25											
26											
27											
28											
29											
30											
31											

Legends

1	Status flag is copied unchanged
-1	Status flag is copied inverted
>0	Status flag is set if at least one of the input flags is True

[sc_Scada-iccp, 1, en_US]

4.9 Mapping of ICCP Quality Flags to Scada Quality Flags

ICCP Quality mapping (ICCP to Scada)													
ICCP Quality Flag											Scada Quality Flags		
											Bit	Quality Flag	Meaning
Validity				Source				Normal Value		other			
VALID	HELD	SUSPECT	NOT VALID	TELEMETERED	CALCULATED	ENTERED	ESTIMATED	NORMAL	ABNORMAL	NOT UPDATED			
	1										0	Telemetry blocked local	yes/ no
-1										1	1	Telemetry blocked remote	yes/ no
											2	Not Updated	yes/ no
			1								3	Invalid local	yes/ no
											4	Invalid remote	yes/ no
		1									5	Suspect local	yes/ no
											6	Suspect remote	yes/ no
											7	Selection Manually Adjusted	yes/ no
											8, 9	Source Selection	00: Primary 01: Secondary 10: Tertiary 11: Auxilliary
				1	1		1				10, 11	Value Source	00: Telemetered 01: Calculated 10: Estimated 11: Entered
				1									Value Source = Telemetered 00: Normal 01: Spontaneous 10: General Interrogation 11: Remote Control Response
					1								Value Source = Calculated 00: Operands valid 01: Operand(s) manually entered 10: Operand(s) suspect 11: Operand(s) invalid
							1						Value Source = Estimated Unobservable yes/ no Bad data yes/ no
								1					Value Source = Entered 0: Local 1: Remote
						1							
											14	True Time	yes/ no
											15	Object in test	yes/ no
											16	Clapped to Zero	yes/ no
											17	Initial Data	yes/ no
								-1			18	Not in normal state	yes/ no
											19	Calculated by RAP auth. processing	yes/ no
											20, 21	RAP authorization source	00: Main 01: Control 10: Integral 1 11: Integral 2
											22	Incomplete aggregation	yes/ no
											23	Alarm suppress	yes/ no
											24	Free bits for future use	
											25		
											26		
											27		
											28	Spare bits reserved for	
											29	project enhancements	
											30		
											31		

Legends

1	Status flag is copied unchanged
-1	Status flag is copied inverted

[sc_iccp-scada, 1, en_US]